

DH2323 Project Specification - Version 1

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Course: DH2323 Computer Graphics and Interaction, KTH

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Intended Grade: Aiming for the highest grade

Project Title

Charlie / Chat - A Browser-Based Cinematic Pixel Lighting Engine

Concept

Charlie / Chat is a short interactive horror vignette built on top of a custom real-time lighting and shadow engine, running in the browser via p5.js and GLSL fragment shaders. The core idea is that the renderer itself produces the horror.

The player controls **Charlie**, a pixel-art character navigating three rooms. The antagonist, **Charlie Chat**, is not a traditional sprite - it is a white silhouette whose visual presence is entirely a product of the SDF shadow and lighting system, growing more opaque and spatially dominant as the threat level rises. The tone is uncanny comedy-horror: Charlie Chat communicates in cheerful sparkle emojis, and that cheerfulness is the dread.

The story is based on an original narrative called *Let Me Ask Charlie*, reframed into a horror context. A single binary player choice (YES / NO) creates two emotional outcomes that converge at the same ending.

Technical Direction

Technical contribution: a **Normal-Mapped Sprite Pipeline in the browser**:

- 2D pixel-art sprites carry a companion normal map, allowing dynamic point lights to produce physically-plausible N·L diffuse shading
- Charlie Chat's form is rendered using an **SDF-based soft shadow raymarcher**
- A post-processing stack handles the cinematic aesthetic: bloom, vignette, chromatic aberration, film grain, and a matrix-based colour grade
- Audio is spatial: Charlie Chat's ambient drone is panned to its screen position using the Web Audio API

Visual Reference

Saul Leiter and Ernst Haas: street photography characterised by desaturated colour, light bleeding through fog, and figures half-consumed by shadow.

Literature I Will Be Reading

- **Inigo Quilez**: 2D SDF reference and raymarching articles (iquilezles.org) - technical foundation for the shadow system
- **Real-Time Rendering, Ch. 9** (Haines et al.): physically-based shading, N·L diffuse model
- **GPU Gems Vol. 1, Ch. 13**: soft shadow techniques, - grounding
- **Reinhard et al. (2002)**: tone mapping - for the color grade pass
- **Heikkinen & Holopainen**: player perception study - foundation for the user study design

Current Status

No code written yet.

GitHub repository: <https://github.com/ashmitroy/dh2323-charlie-chat>

Blog: <https://ashmitroy.github.io/dh2323-charlie-chat/>

Questions for the Course Team

1. Would a p5.js + GLSL browser implementation be treated equivalently to a native OpenGL project for grading purposes?
2. Is a three-room interactive vignette sufficient scope for the application layer, given the rendering system is the primary contribution?
3. Any advice on normal map pipeline for 2D pixel art specifically: hand-painted maps vs. procedural derivation?

This is a first draft. The story details, character design, and technical implementation plan will be added in subsequent versions.